

PHASE 4:- DOM Manipulation

#DOM Selection and Manipulation

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JS can use the DOM to -

- select elements ✓
- change text, style ✓
- Add / Remove elements
- handle events.

Selecting Element -

JS DOM Element Selection Methods Different ways to find elements in the DOM using JavaScript				
Method	What it selects	Returns	Example	Multiple elements?
✓ <code>getElementById()</code> ✓	Element with a specific ID	Single element	<code>document.getElementById("title")</code>	✗ No
✓ <code>getElementsByClassName()</code>	Elements with a specific class	HTMLCollection	<code>document.getElementsByClassName("box")</code>	✓ Yes
✓ <code>getElementsByTagName()</code>	Elements with a specific tag	HTMLCollection	<code>document.getElementsByTagName("p")</code>	✓ Yes
✓ <code>querySelector()</code>	First element matching a CSS selector	Single element	<code>document.querySelector(".box")</code>	✗ Only first
✓ <code>querySelectorAll()</code>	All elements matching a CSS selector	NodeList	<code>document.querySelectorAll(".box")</code>	✓ Yes

💡 Quick Way to Remember

Method	Selector style	Example
<code>getElementById()</code>	ID	<code>#title</code> → <code>"title"</code>
<code>getElementsByClassName()</code>	Class	<code>.box</code> → <code>"box"</code>
<code>getElementsByTagName()</code>	Tag	<code>p</code> → <code>"p"</code>
<code>querySelector()</code>	CSS selector	<code>".box", "#title", "p"</code>
<code>querySelectorAll()</code>	CSS selector	<code>".box", "#title", "p"</code>

★ Important Difference

`querySelector()` and `querySelectorAll()` are more flexible because they support CSS selectors such as:

- **.class** (e.g. `.box`)
- **#id** (e.g. `#title`)
- **tag** (e.g. `p`)
- **combined** (e.g. `div.box`)
- **attribute** (e.g. `[type="text"]`)

```
// changing text
// textContent
// const title = document.getElementById("title")
// console.log(title.textContent)
// title.textContent = "Satyam kumar" //changing text of element
// console.log(title.textContent)

// innerHTML

// const title = document.getElementById("title")
// console.log(title.innerHTML)
// title.innerHTML = "<span>Satyam kumar</span>" //changing text of element
// console.log(title.innerHTML)
```

```

// changing styles
// const title = document.getElementById("title")
// // console.log(title.style)
// title.style.backgroundColor = "red"

// working with classes
// .classList
// add class
// const title = document.getElementById("btn")
// // console.log(title.classList)
// title.classList.add("welcome")
// // console.log(title.classList)
// // remove class
// title.classList.remove("text")
// // console.log(title.classList)
// // toggle classlist
// title.addEventListener("click", ()=>{
//     title.classList.toggle("text")
// })
// attributes
// const title = document.getElementById("title")
// // console.log(title.getAttribute("class"))
// // title.setAttribute("class","domclass")
// title.removeAttribute("id")

// create element
// const domH1 = document.createElement("p")
// domH1.textContent = "Dynamic set value"
// domH1.style.color = "red"
// domH1.classList.add("ramu")

// // append() -> insert item/elements at the last child of the body
// document.body.appendChild(domH1)
// document.body.appendChild(domH1, "Extra content")
// appendChild()
// document.body.appendChild(domH1, "Extra")

// remove element
// const title = document.getElementById("title")
// title.remove()

```

mini - project

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>

  <h1 id="title">My website</h1>
  <button id="btn">Change title</button>

  <script>
    const title = document.getElementById("title")
    const btn = document.getElementById("btn")
    btn.addEventListener("click", ()=>{
      title.textContent = "Welcome Satyam"
      title.style.color = "red"
    })
  </script>
</body>
</html>
```